

Implementing Mixed-Data Sampling Models for Temporal Sampling and Aggregation in Stata

`mfcollapse` and `bumidas`

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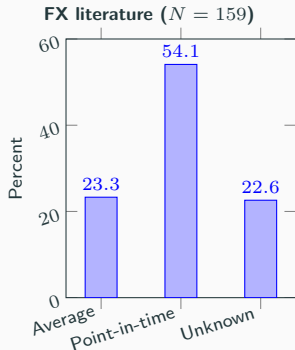
Wilfrid Laurier University

Three recent insights about temporal aggregation

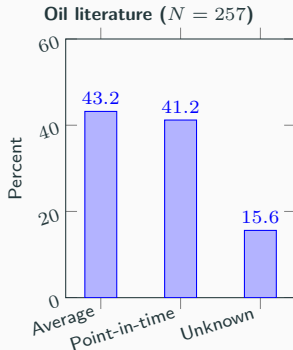
1. Forecasts of temporal aggregates (real oil prices, REERs) **failed to test the random walk null** — the end-of-period no-change — resulting in **spurious predictability**.
2. Bottom-up high-frequency forecasting methods can **reduce mean squared forecast error by up to 54%** using daily data relative to forecasts using only lower-frequency data.
3. Structural models estimated with temporally aggregated data can **mistime shocks by up to 70%** and introduce **spurious endogeneity** in VAR and DSGE models.

Temporally aggregated data belong on the LHS, not the RHS, of equations.

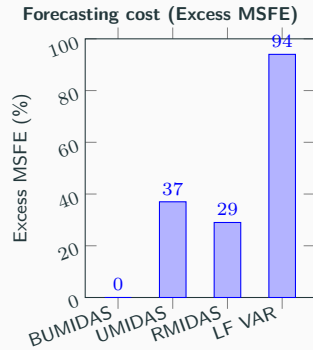
Implications for Forecasting Literature



Share of exchange-rate papers by sampling

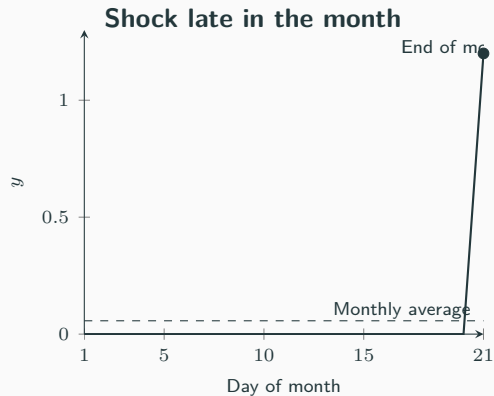
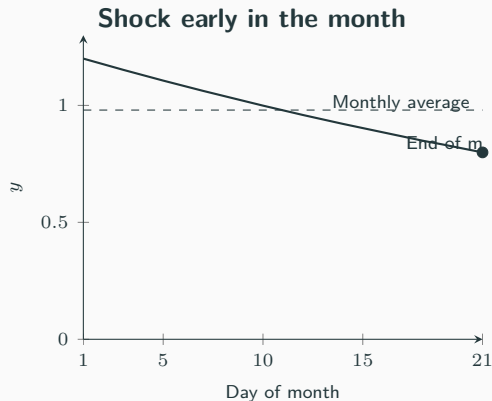


Share of oil-price papers by sampling



relative to bottom-up: VAR(1) sims
 $n = 20$, $T = 200$

Why aggregation loses information: a persistent AR(1)



Same persistence, very different information at the forecast origin.

Aggregation changes coefficients, not the state dimension

Suppose a monthly target is the average of a daily AR(1):

$$(1 - \rho L)y_{t,i} = \varepsilon_{t,i}, \quad \bar{y}_t = \frac{1}{n} \sum_{i=1}^n y_{t,i}.$$

Brewer (1973) Strategy

Aggregate representation:

$$\bar{y}_{T+h} = (1 + (1+\rho)L + \dots + \rho^{n-1}L^{2(n-1)})\varepsilon_{t,i}.$$

Many coefficients, and their number increases with n .

Bottom-up Strategy

Optimal forecast representation:

$$\mathbb{E}_{T,n}[\bar{y}_{T+h}] = \frac{1}{n} \sum_{i=1}^n \rho^{(h-1)n+i} y_{T,n} = \beta_h y_{T,n}.$$

One state is sufficient, regardless of n .

Recursive and direct bottom-up forecasts project on the same high-frequency state

BUMIDAS: general VARMA(p, q)

For a high-frequency VARMA(p, q),

$$\mathbb{E}_{T,n} [\bar{Z}_{T+h}] = \sum_{j=1}^p \beta_{h,j} Z_{T,n-j+1} + \sum_{\ell=1}^q \alpha_{h,\ell} \varepsilon_{T,n-\ell+1}$$

- **Parsimonious:** latest p AR and q MA terms only
- **Invariant:** state dimension does not grow with n
- **Data requirement:** high-frequency target history observed or recoverable
- **Optimal:** direct BUMIDAS = recursive bottom-up

Four new Stata commands for mixed-frequency forecasting



Construct

Daily, weekly, or monthly data
Preserve the relevant high-frequency history

Estimate

Estimate BUMIDAS directly
Select the high-frequency state by IC

Companion commands `umidas` and `rmidas` estimate a suite of unrestricted and restricted MIDAS models.

What does mixed-frequency data look like in Stata?

Data Editor

t	i	$y_{t,i}$
1	1	$y_{1,1}$
1	2	$y_{1,2}$
1	3	$y_{1,3}$
2	1	$y_{2,1}$
2	2	$y_{2,2}$
2	3	$y_{2,3}$
3	1	$y_{3,1}$
3	2	$y_{3,2}$
3	3	$y_{3,3}$

High-frequency data

$n = 3$

$t = \text{period}$

$i = \text{observation within period}$

Command

```
list t i y
```

What does mixed-frequency data look like in Stata?

Data Editor

t	$\bar{y}_t$	$y_{t,n}$
1	$\bar{y}_1$	$y_{1,3}$
2	$\bar{y}_2$	$y_{2,3}$
3	$\bar{y}_3$	$y_{3,3}$
$\vdots$	$\vdots$	$\vdots$

Average: $\bar{y}_t$

End of period: $y_{t,n}$

MIDAS: $\bar{y}_t = \alpha + \beta y_{t-1,n} + u_t$

Command

```
collapse (mean) ybar=y  
        (last) yn=y, by(t)
```

```
regress ybar L.yn
```

What does mixed-frequency data look like in Stata?

Data Editor

t	$\bar{y}_t$	$y_{t,n}$	$y_{t,n-1}$	$y_{t,n-2}$
1	$\bar{y}_1$	$y_{1,3}$	$y_{1,2}$	$y_{1,1}$
2	$\bar{y}_2$	$y_{2,3}$	$y_{2,2}$	$y_{2,1}$
3	$\bar{y}_3$	$y_{3,3}$	$y_{3,2}$	$y_{3,1}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Average: $\bar{y}_t$

HF state: $y_{t,n}, y_{t,n-1}, y_{t,n-2}$

MIDAS:
$$\bar{y}_t = \alpha + \beta_0 y_{t-1,n} + \beta_1 y_{t-1,n-1} + \beta_2 y_{t-1,n-2} + u_t$$

Command

```
mfcollapse y, ...  
lags(2)
```

```
regress ybar L.yn L.yn1 L.yn2
```

mfcollapse: construct mixed-frequency data

Syntax

```
mfcollapse varlist, date(varname) frequency(string) [ar(#) lags(#)  
anchor(varname)]
```

Example

```
mfcollapse wti, date(time)  
frequency(monthly) ar(1)
```

Creates

```
wti_ld0    wti_ld1    ...    wti_ldK  
wti_ave
```

Where

- `ld#`: ($t, n - \#$) observation; n inferred automatically
- `ave`: within-period average

Key options

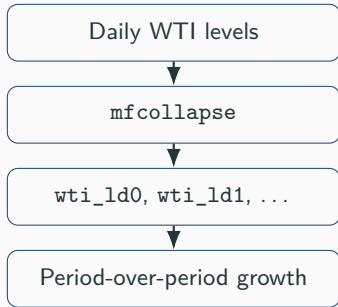
<code>date()</code>	source date; frequency inferred
<code>frequency()</code>	lower-frequency target
<code>ar(p)</code>	$K = p(n - 1)$; default $p = 1$
<code>lags(k)</code>	set $K = k$; default calculated
<code>anchor()</code>	common observation schedule

Supported aggregation paths

daily	→	weekly, monthly, quarterly
weekly	→	monthly, quarterly
monthly	→	quarterly

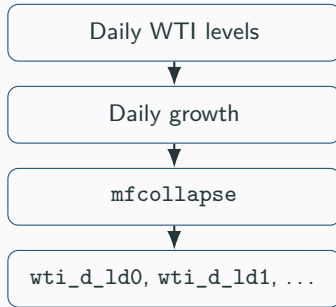
Transform before or after mfcollapse?

Transform after



Change across HF periods

Transform before



Change across HF observations

mfcollapse is transformation-neutral: the two specifications contain different information.

Different observation schedules are an advantage of mixed-frequency data

Observed high-frequency data

Date	FX	Oil	Gas
Mon	y_1	—	—
Tue	y_2	x_1	—
Wed	—	x_2	—
Thu	y_3	x_3	—
Fri	y_4	x_4	g_1
Mon	y_5	x_5	—
Tue	y_6	x_6	—
Wed	y_7	x_7	—
Thu	—	x_8	—
Fri	y_8	—	g_2

- Oil/ FX: some missing days
- Gas: Fridays

Weekly mixed-frequency data after mfcollapse

Series	ld0	ld1	ld2	ld3
FX	y_4	y_3	y_2	y_1
Oil	x_4	x_3	x_2	x_1
Gas	g_1			
FX	y_8	y_7	y_6	y_5
Oil	x_8	x_7	x_6	x_5
Gas	g_2			

- No need to align to the same daily calendar
- No need to drop to get common dates
- Daily and weekly series can be combined

How does BUMIDAS work?

Think `auto.arima` for MF direct forecasts: IC selects unrestricted HF/LF lag order

1. Use the richest information available

- **HF target observed:** use HF history of target—information usually omitted from standard MIDAS.
- **Credible HF proxy:** use the proxy history to recover information about the latent HF target.
- **HF target unavailable:** use the standard MF information set: LF target history and available HF predictors.

2. Select an unrestricted model

- **Like UMIDAS:** each retained HF observation has its own coefficient.
- **Bottom-up parsimony:** HF depends on p, q of DGP, not degree of aggregation.
- **Unknown p, q :** IC selects unrestricted HF/LF lag orders
- **Signal extraction:** when the HF target is noisy or unavailable, extract the signal from both HF and LF dynamics.

Gains from better information + fewer unnecessary parameters

bumidas: estimate and select the direct forecast

Syntax

```
bumidas depvar, [hftarget(stublist) hfpredictors(stublist) hfmax(#)
             search(full|recursive|none) ic(bic|hqic|aic) horizon(#) ...]
```

Example

```
bumidas y_ave, hftarget(y_ld)
         hfpredictors(wti_ld) hfmax(20)
         search(recursive) ic(bic)
```

Selects

HF order for each block and the final direct BUMIDAS regression.

Postestimation

```
predict double yhat
```

Key options

hftarget()	required HF blocks
hfpredictors()	optional HF blocks
hfmax()	maximum HF order
search()	full / recursive / none
ic()	BIC / HQIC / AIC
horizon()	direct forecast horizon

Also supports

LF target lags, LF predictors, controls, and fixed orders.

How does bumidas select lag orders?

`search(full)`

All admissible
block-order combinations



Lowest information
criterion

Global IC minimum

`search(recursive)`

Start from minimum
admissible model



Add one HF term
at a time



Stop when IC
no longer improves

Fast greedy search

`search(none)`

User specifies
block orders



Estimate fixed
BUMIDAS model

Prespecified orders

All candidate models are estimated on the same sample.

Default: `search(full), ic(bic), horizon(1)`

1. Provide high-frequency lags of target/proxies and predictors
2. Estimate BUMIDAS
3. Generate the direct forecast

Mixed-frequency data → selected direct forecast

BUMIDAS recovers the optimal bottom-up forecast

Method	Rel. MSFE	MDA
High Frequency		
Bottom-up recursive	0.489	0.720
MIDAS		
BUMIDAS	0.476	0.740
BUMIDAS-BIC	0.476	0.740
UMIDAS	0.669	0.680
RMIDAS	0.632	0.640
Lower frequency / no-change		
LF VAR	0.949	0.620
EoP no-change	0.677	0.680
LF no-change	1.000	0.000

VAR(1) Simulations

$$n = 20$$

$$T = 200$$

BUMIDAS has lowest MSFE

**BUMIDAS-BIC works even
when DGP unknown**

Large gains over LF VAR

Daily information delivers large out-of-sample gains

One-month-ahead FX forecast

Method	Rel. MSFE	MDA
EoP no-change	0.660	0.690
BUMIDAS	0.660	0.680
LF VAR	0.930	0.580
UMIDAS	0.960	0.600
RMIDAS	0.920	0.570

30+% lower MSFE

when preserving high-frequency
data

Relative MSFE against the period-average no-change benchmark.

UMIDAS and RMIDAS use the companion Stata commands.

The first-order gain comes from using the high-frequency target information.

Install MIDAS for Stata

Install directly from GitHub

```
net install bumidas,  
    from("https://raw.githubusercontent.com/  
        SSEconomics/bumidas-stata/main")  
replace
```

Documentation

```
help mfcollapse  
help bumidas  
help umidas  
help rmidas
```

Includes

- Almon, step, Legendre, exponential, and beta
- Bottom-up recursive methods
- Simulation code and applications



github.com/SSEconomics/bumidas-stata

Three things to remember

1. Stata can easily handle high-frequency and MIDAS models
2. High-frequency data is more valuable than previously realized
3. Keep temporally aggregated targets on the LHS—use high-frequency data on the RHS

`github.com/SSEconomics/bumidas-stata`